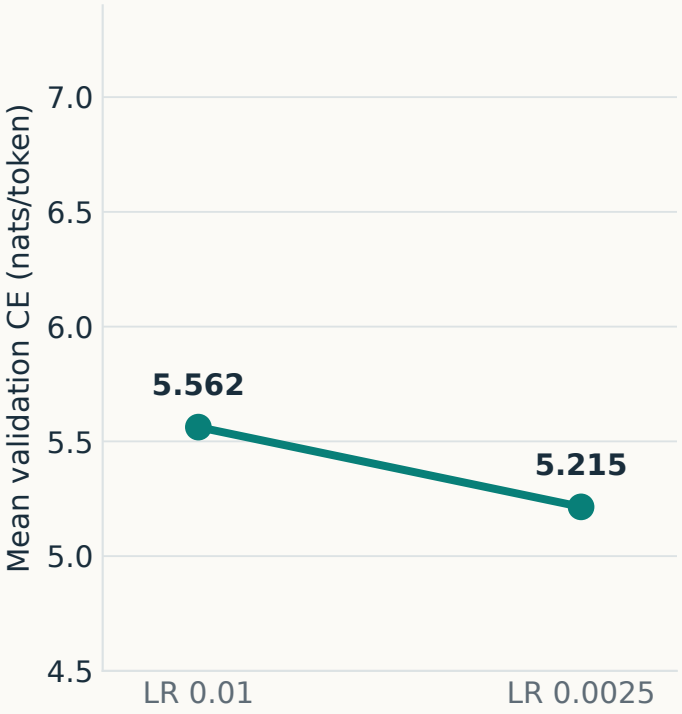
The largest early gains came from the training recipe

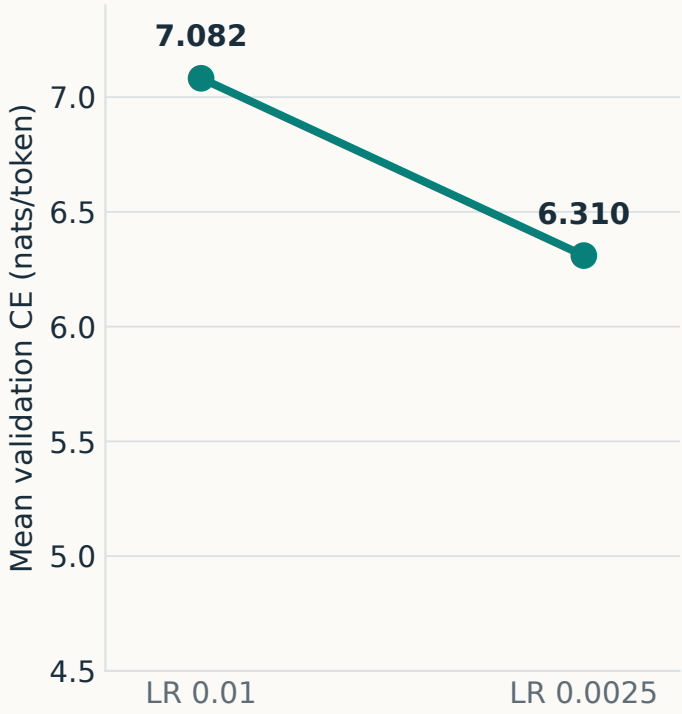
Three fresh paired seeds per comparison · GPT-2 BPE on Tiny Shakespeare · lower validation CE is better

MHA · width 64 · 3,000 updates



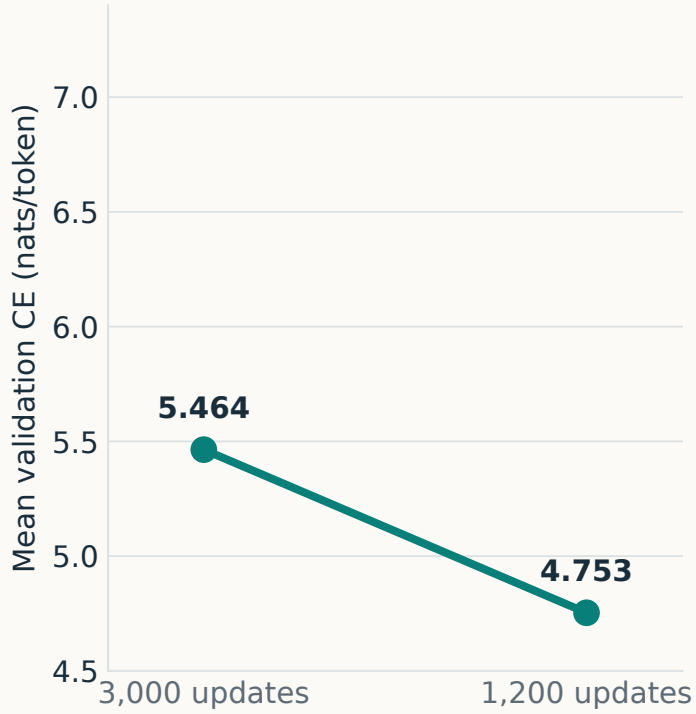
Δ CE = -0.347

MHA · width 256 · 3,000 updates



Δ CE = -0.773

GQA · width 256 · LR 0.00125



Δ CE = -0.711

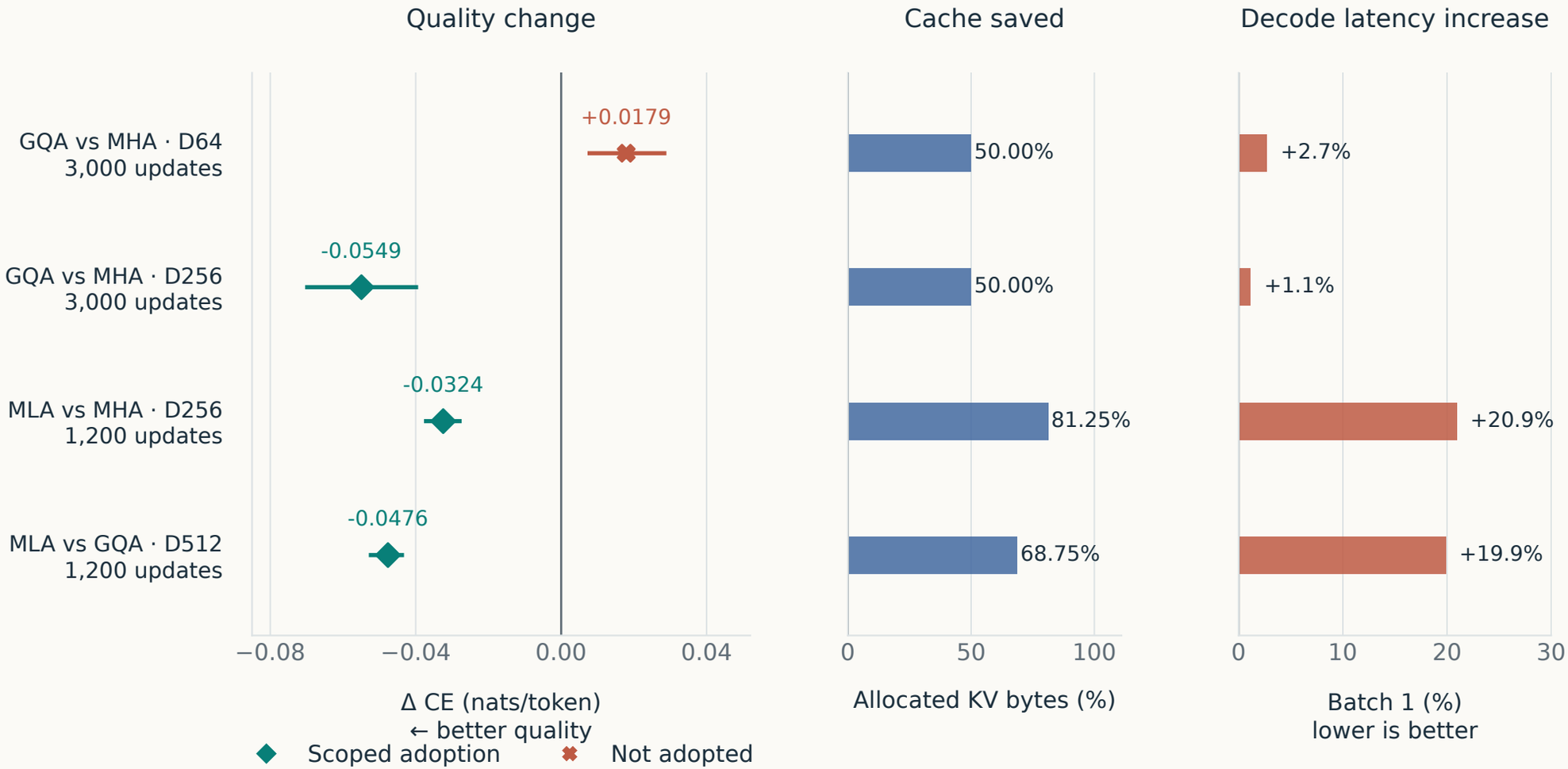
Training time: 504 → 202 s (−60%)

Separate controlled comparisons; do not add their gains. Horizon, cooldown and EMA start changed together.

H200 = EMA half-life 200 updates. Means use reused validation text; test holdout remains unscored. Sources: quality-001; horizon-001.

A smaller attention cache did not mean faster decoding

Each row is a separate fresh comparison; GQA's quality result also changed with model width

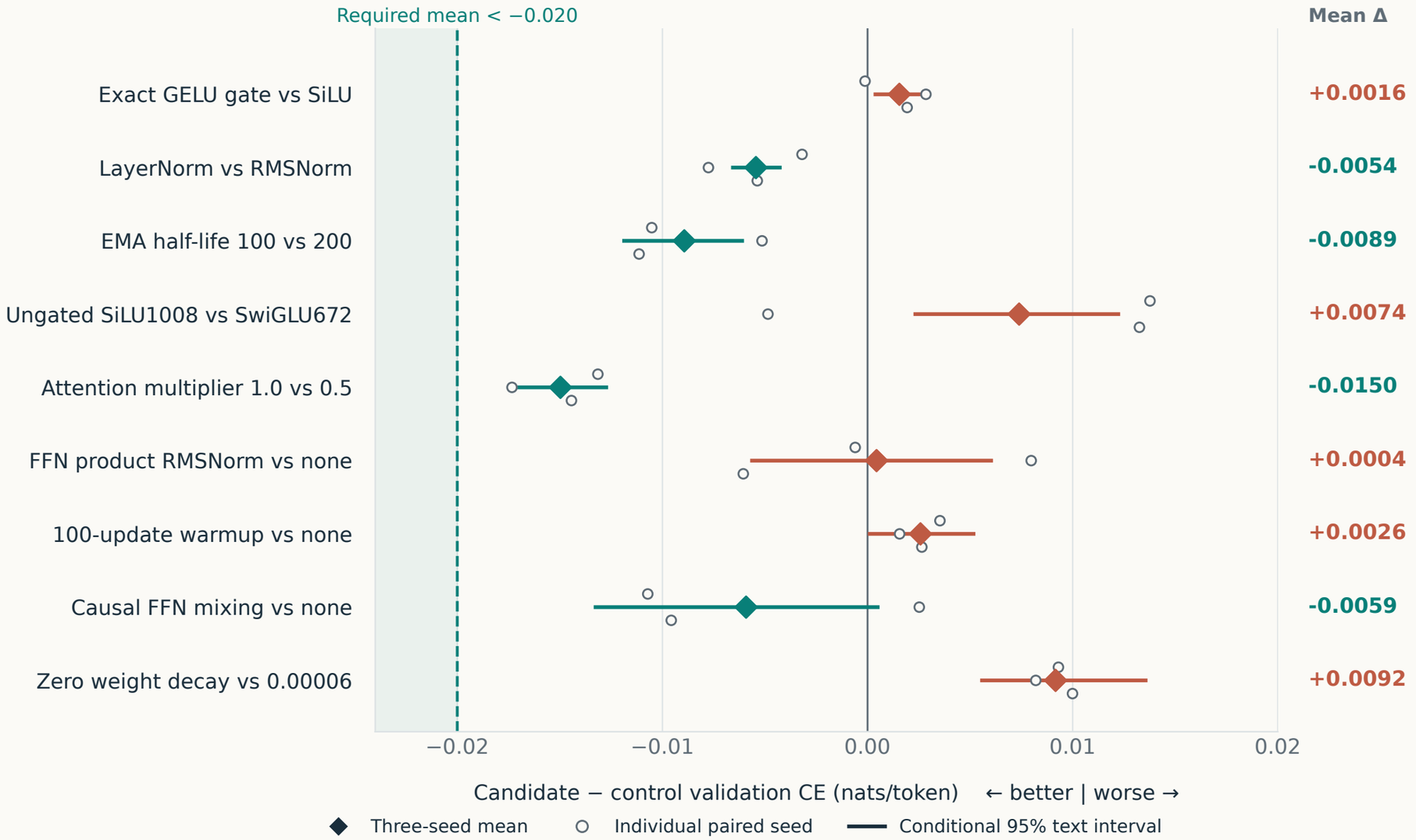


Whiskers: conditional 95% text-block intervals (4,096-token blocks), not seed-population uncertainty.

FP32 MPS, context 128, 32-token prefill / 64-token decode; two models resident. Cache ≠ total memory. Sources: GQA, MLA, wide-attention final reviews.

Later refinements found small gains, but no new adoption

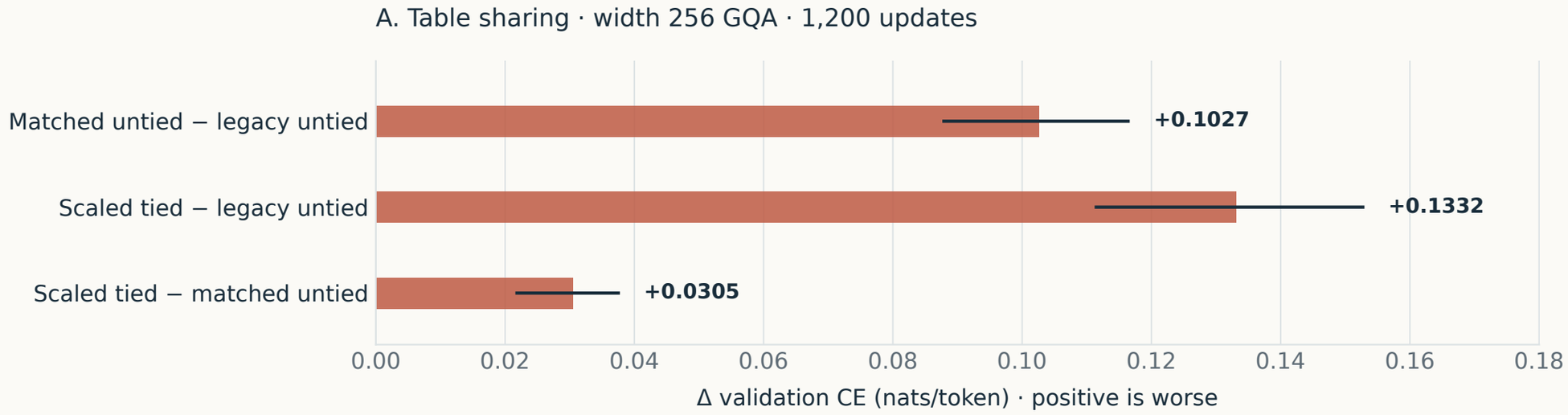
Width 256 RoPE MLA · 1,200 updates · nine completed follow-ups, shown in research order



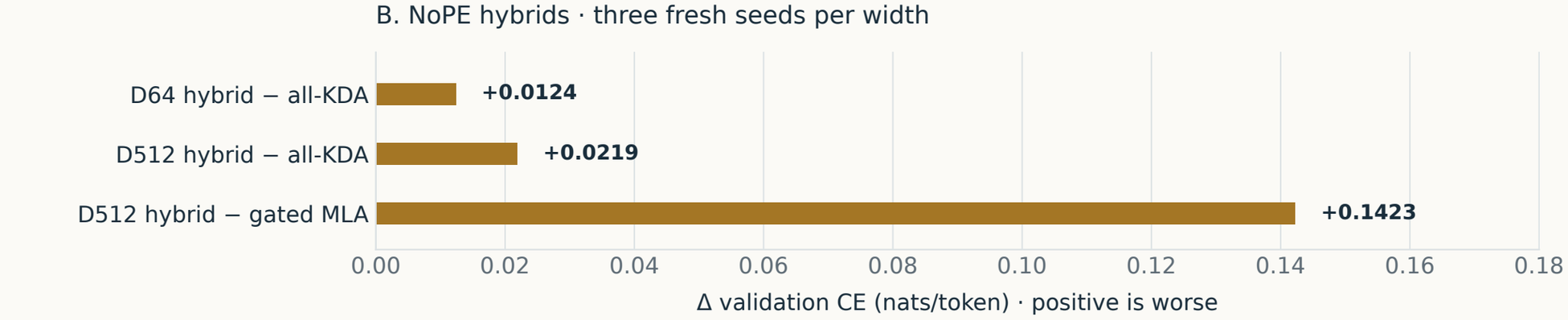
Attention scale, EMA half-life and LayerNorm improved all three seeds, but missed the predeclared minimum gain. No effects are added together.
Whiskers use 4,096-token blocks; adoption also required every seed and all simultaneous text bounds to improve. Intervals are not holdout confidence.
Equal per-arm LR searches where applicable; EMA uses the same online trajectory, and decay uses fixed common LR 0.00125. Sources: nine final quality reviews.

Compression and hybridization still had to pass quality gates

Neither vocabulary-table sharing nor the tested 3:1 KDA / gated-MLA hybrids earned adoption



45.0% less parameter storage; +0.1332 CE vs legacy. Only +0.0305 is the matched tying contrast.



D512 hybrid also used 21.9% more state than gated MLA and 18.9% more than KDA at context 128. These recipes are local adaptations.

Panel A whiskers: conditional 95% text intervals; Panel B: means only. No general rejection of tying/hybrids. Sources: table-001; hybrid-001; wider-hybrid closure.